

Strategic Insights on Speed Dating

The entrepreneur wants to optimally match couples on a dating app. To help, we first check the dataset for import errors, then build equations predicting how much a dater likes their partner based on attractiveness, sincerity, intelligence, fun, ambitiousness, and shared interests. We will build separate models for both males and females and assess if they agree with respect to any of the characteristics of interest. Whether shared race or age proximity play a role in liking will also be examined.

The importance here comes from understanding what attributes each gender values most. This will allow us to predict ideal partner matches, in turn increasing user satisfaction, and therefore delivering strong returns for the entrepreneur and his investors.

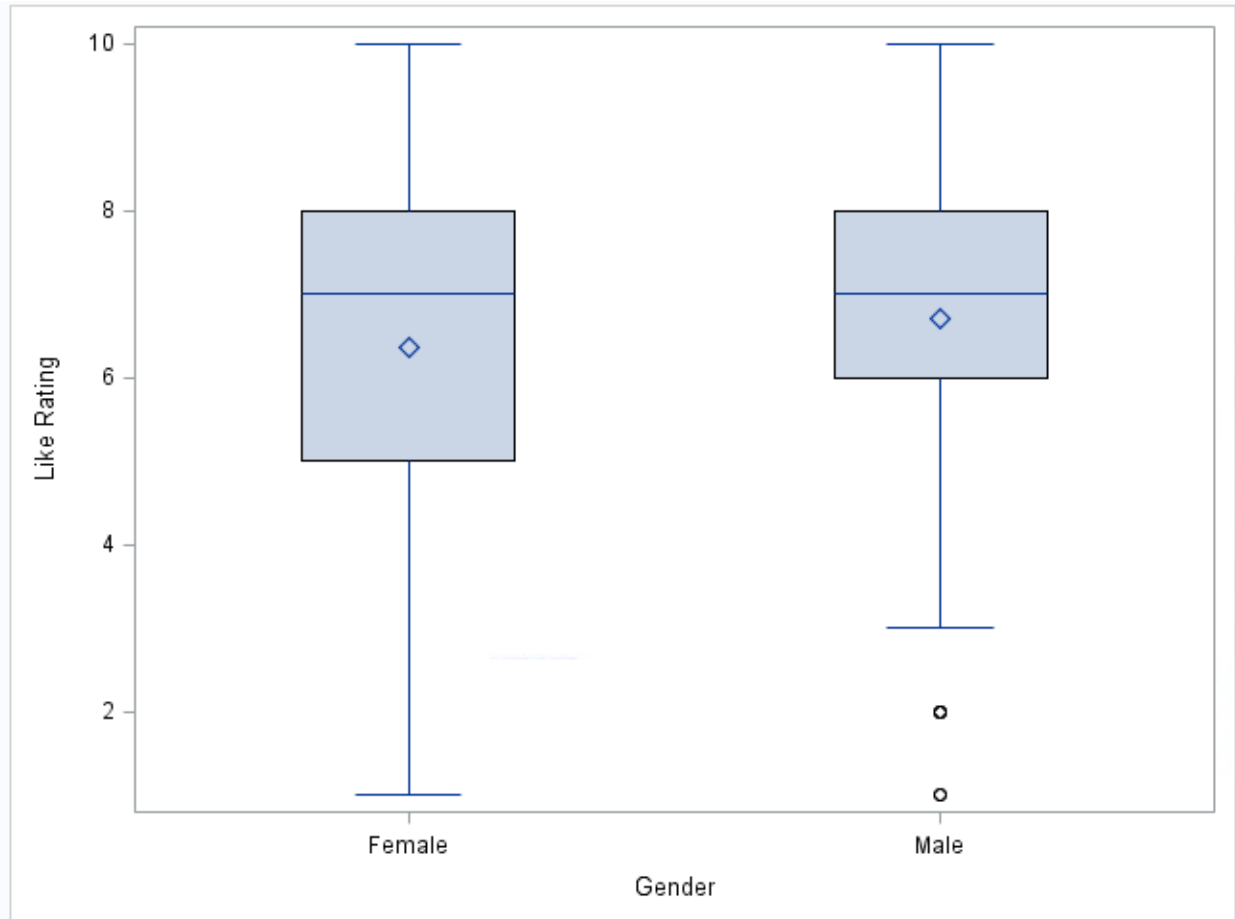
The following variables included in this study are ordinal: Attractive, Sincere, Intelligent, Fun, Ambitious, SharedInterests. We treat them as numerical for modeling as they are on a scale 1-10, 1=awful, 10=great. Our response is Like (1=don't like at all, 10= like a lot). Other numerical predictors include Age (in years) and PartnerYes (how probable do you think it is that this person will say 'yes' for you? 1=not probable, 10=extremely probable). Categorical variables are Race (Caucasian, Asian, Black, Latino, or Other), and Decision (1=Yes, I want to see the date again, 0=No I don't want to see the date again). It's important to note that variables like DecisionM refers to the Male's decision regarding whether he wants to see the female again. LikeM refers to how much the male liked the female. AttractiveM refers to the male's rating of the female's attractiveness. However, AgeM and RaceM refers to the male's age and race. Also, the variables Decision and PartnerYes are included in the variable descriptions given by the entrepreneur, but not actually included in the data set sent over. Data entry errors include the

SharedInterests and Fun variables having observations with a rating of 0, which is not included on the scale. These observations were removed from the data set.

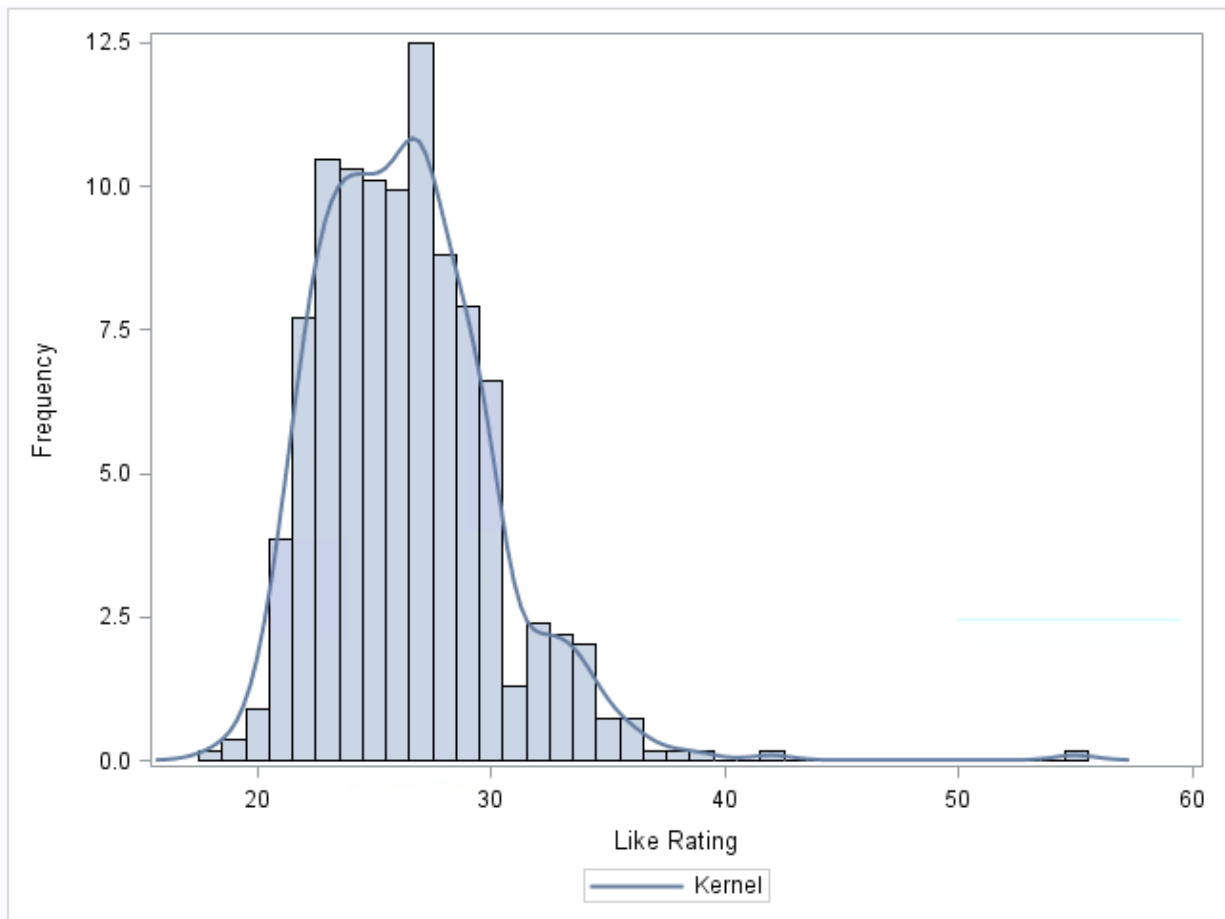
Starting with model reliability – how well our model performs when predicting future values – the male model, for predicting how much a male would like their female partner, yields a shrinkage of 0.0566 and the female model with 0.0088. As they are both well below the 0.10 threshold, both models hold up when working with new individuals.

Let's take a look at some descriptive statistics of our response, Like. Based on 268 and 266 recorded observations respectively, both males and females had a median liking rating of 7 for their partner. The mode is also 7 for both, with each gender's ratings ranging from 1 to 10, a range of 9. The average rating males gave their female dates is 6.71, with a standard deviation of ± 1.78 . However, the average rating females gave their male dates is 6.37, with a standard deviation of ± 1.76 . This means that males rated their female counterparts both higher on average and had more variation in their ratings. No correlation coefficients had an absolute value of 0.95 or higher, so collinearity is not a concern, although we will dive more in depth in later sections.

The boxplot below represents the descriptive distribution of the variables LikeM and LikeF in the dataset.



The distribution of ages of all individuals can be seen in the figure below.



The histogram is skewed to the right, with looks to be 2 separate peaks, suggesting the presence of two subgroups in the data. This may be a result of the distributions of male ages and female ages differing.

The entrepreneur was also interested in how many couples had the same race and how many did not. This can be seen in the table below.

SameRace	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	144	54.55	144	54.55
1	120	45.45	264	100.00
Frequency Missing = 6				

Of the 264 couples with complete race data, 120 were the same race and 144 were not. Six observations were excluded due to missing values in RaceM or RaceF. He was also interested in how many couples are "close" in age and how many are not, defined as by being within 2 years of one another, which can be seen in the table below.

CloseAge	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	154	58.78	154	58.78
1	108	41.22	262	100.00
Frequency Missing = 8				

There are 108 couples that are close in age, and 154 couples that are not. Note that 8 observations were excluded due to missing age for at least one of the variables, AgeM or AgeF.

For variable selection in predicting how speed daters liked their partners, we will use 4 strategies with a significance of .05. The first being all possible models, where our most optimal model will be when $C(p) < \text{number in model}$. The “Male Model” figure is the model that predicts how much males like females.

Male Model

Number in Model	R-Square	C(p)	MSE	Variables in Model
4	0.3137	210.5926	2.22549	SincereM IntelligentM SameRace CloseAge
4	0.2763	233.7083	2.34694	IntelligentM AmbitiousM SameRace CloseAge
5	0.6502	4.9279	1.13968	AttractiveM SincereM FunM AmbitiousM SharedInterestsM
5	0.6489	5.7314	1.14392	AttractiveM SincereM FunM SharedInterestsM SameRace
5	0.6471	6.8391	1.14977	AttractiveM SincereM IntelligentM FunM SharedInterestsM
5	0.6467	7.0592	1.15093	AttractiveM SincereM FunM SharedInterestsM CloseAge

The optimal male model has 5 variables. The variables are AttractiveM, SincereM, FunM, AmbitiousM, and SharedInterestsM.

Female Model

Number in Model	R-Square	C(p)	MSE	Variables in Model
4	0.2071	240.9841	2.52883	SincereF AmbitiousF SameRace CloseAge
5	0.6224	4.6581	1.20972	AttractiveF SincereF IntelligentF FunF SharedInterestsF
5	0.6212	5.3968	1.21385	AttractiveF IntelligentF FunF SharedInterestsF SameRace
5	0.6197	6.2125	1.21840	AttractiveF IntelligentF FunF SharedInterestsF CloseAge
5	0.6192	6.5098	1.22006	AttractiveF IntelligentF FunF AmbitiousF SharedInterestsF
5	0.6119	10.7336	1.24365	AttractiveF SincereF FunF SharedInterestsF SameRace

The optimal female model has 5 variables. The variables are AttractiveF, SincereF, IntelligentF, FunF, and SharedInterestsF.

Next, we will use forward selection, which means we sequentially add significant terms starting from no terms, seen in the figures below.

Male Model

Summary of Forward Selection							
Step	Variable Entered	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	AttractiveM	1	0.5573	0.5573	54.2612	278.20	<.0001
2	FunM	2	0.0464	0.6037	27.6468	25.73	<.0001
3	SharedInterestsM	3	0.0281	0.6317	12.3183	16.69	<.0001
4	SincereM	4	0.0145	0.6463	5.3386	8.97	0.0031

The optimal male model using forward selection has 4 variables, AttractiveM, FunM, SharedInterestsM, SincereM.

Female Model

Summary of Forward Selection							
Step	Variable Entered	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	FunF	1	0.4288	0.4288	107.759	166.68	<.0001
2	AttractiveF	2	0.1098	0.5386	46.7780	52.57	<.0001
3	SharedInterestsF	3	0.0551	0.5937	17.1477	29.85	<.0001
4	IntelligentF	4	0.0255	0.6192	4.5203	14.66	0.0002

The optimal female model using forward selection has 4 variables, FunF, AttractiveF, SharedInterestsF, IntelligentF.

Next we will do backwards elimination, where we start with the maximum model and sequentially remove insignificant terms, seen in the figures below.

Male Model

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	-0.22463	0.41680	0.33320	0.29	0.5905
AttractiveM	0.49939	0.05412	97.68519	85.16	<.0001
SincereM	0.17310	0.05781	10.28485	8.97	0.0031
FunM	0.17635	0.05640	11.21400	9.78	0.0020
SharedInterestsM	0.16525	0.04321	16.77296	14.62	0.0002

Our optimal model has 4 variables, AttractiveM, SincereM, FunM, and SharedInterestsM.

Female Model

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	0.16147	0.42995	0.17131	0.14	0.7076
AttractiveF	0.27828	0.04631	43.85224	36.11	<.0001
IntelligentF	0.21189	0.05534	17.80472	14.66	0.0002
FunF	0.23742	0.05045	26.90262	22.15	<.0001
SharedInterestsF	0.22738	0.04210	35.42173	29.16	<.0001

Our optimal model has 4 variables, AttractiveF, IntelligentF, FunF, and SharedInterestsF.

Lastly, we will do stepwise forward selection, which is the same as forward selection but a variable once found significant can later be removed, as seen in the figures below.

Male Model

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	AttractiveM		1	0.5573	0.5573	54.2612	278.20	<.0001
2	FunM		2	0.0464	0.6037	27.6468	25.73	<.0001
3	SharedInterestsM		3	0.0281	0.6317	12.3183	16.69	<.0001
4	SincereM		4	0.0145	0.6463	5.3386	8.97	0.0031

The optimal model has 4 variables, AttractiveM, FunM, SincereM, and SharedInterestsM. 3 of our approaches zeroed in on the same model – AttractiveM, FunM, SharedInterestsM, and SincereM. Therefore we will use these in our final male model. Note neither SameRace nor CloseAge were selected to be in the final model, in any of the methods, so therefore we can conclude they do not meaningfully contribute to the prediction of how well a male would like their female dating partner.

Female Model

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	FunF		1	0.4288	0.4288	107.759	166.68	<.0001
2	AttractiveF		2	0.1098	0.5386	46.7780	52.57	<.0001
3	SharedInterestsF		3	0.0551	0.5937	17.1477	29.85	<.0001
4	IntelligentF		4	0.0255	0.6192	4.5203	14.66	0.0002

The optimal model has 4 variables, FunF, AttractiveF, SharedInterestsF, and IntelligentF – which 3 of our approaches zeroed in on. Therefore we will use these terms in the final model. Note that neither SameRace nor CloseAge made it to the final model again.

To run diagnostics on our final models, we detect outliers, check the four assumptions, and detect the presence of collinearity. Data entry errors were already addressed earlier. For outliers, observations with a jackknife residual exceeding a magnitude of 2, Cook's distance above 1, or leverage above 0.037 are red flags, as outliers can lead to inaccurate predictions.

Male Model

Female Model

Extreme Observations				Extreme Observations			
Lowest		Highest		Lowest		Highest	
Value	Obs	Value	Obs	Value	Obs	Value	Obs
-3.81605	115	1.94370	158	-3.32099	49	1.90132	216
-3.42787	255	2.04580	254	-2.54646	202	2.09933	91
-3.33439	247	2.15585	252	-2.50992	179	2.64126	96
-3.17992	211	2.22027	222	-2.44404	56	2.72409	1
-2.43562	144	2.44734	32	-2.43800	131	3.61871	250

The jackknife residual plots reveal several observations exceeding our cutoff in both models

Male Model

Female Model

Obs	cooks	Obs	cooks
1	0.23183	1	0.076736
2	0.10097	2	0.061743
3	0.05689	3	0.060582
4	0.05448	4	0.057882
5	0.05203	5	0.054265

Cook's D, which measures an observation's overall influence, yielded no violations.

Lastly, leverage, measures couples in the dataset who have an extreme or unusual combination of characteristics. Leverage can be seen in the figures below.

Male Model

Obs	leverage
1	0.12290
2	0.09592
3	0.08184
4	0.07771
5	0.06115

Female Model

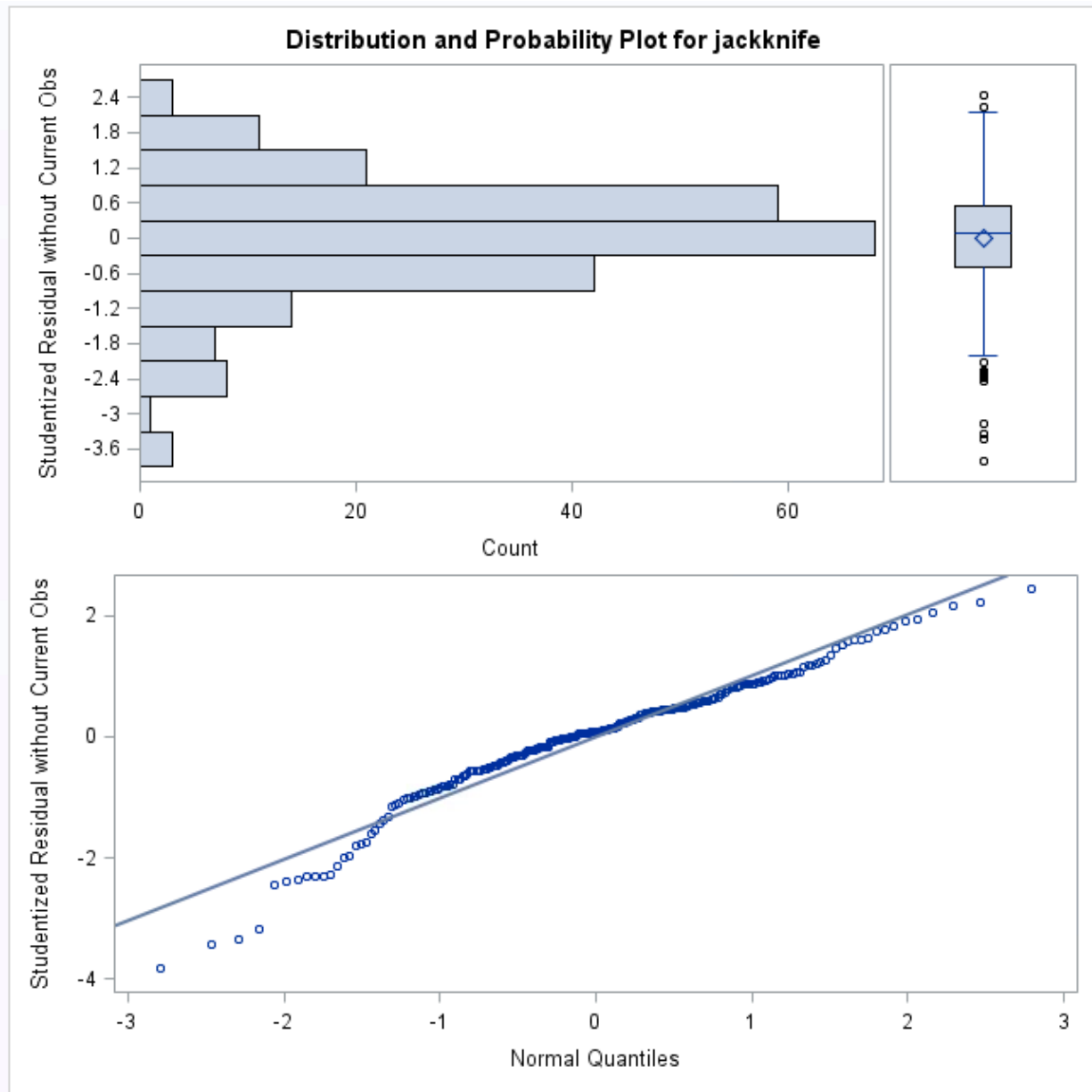
Obs	leverage
1	0.081175
2	0.078239
3	0.070771
4	0.069156
5	0.065425

Again, multiple violations of our cutoff can be seen here. A follow up examination of the data found multiple outliers among the AgeM and AgeF variables. However, as the observations are not data entry errors, part of the intended population, and are not unusual, we keep them.

Next, we have to test the residuals for normality, homogeneity, if they have a mean of zero, and independence. If these assumptions are violated, we cannot draw proper conclusions.

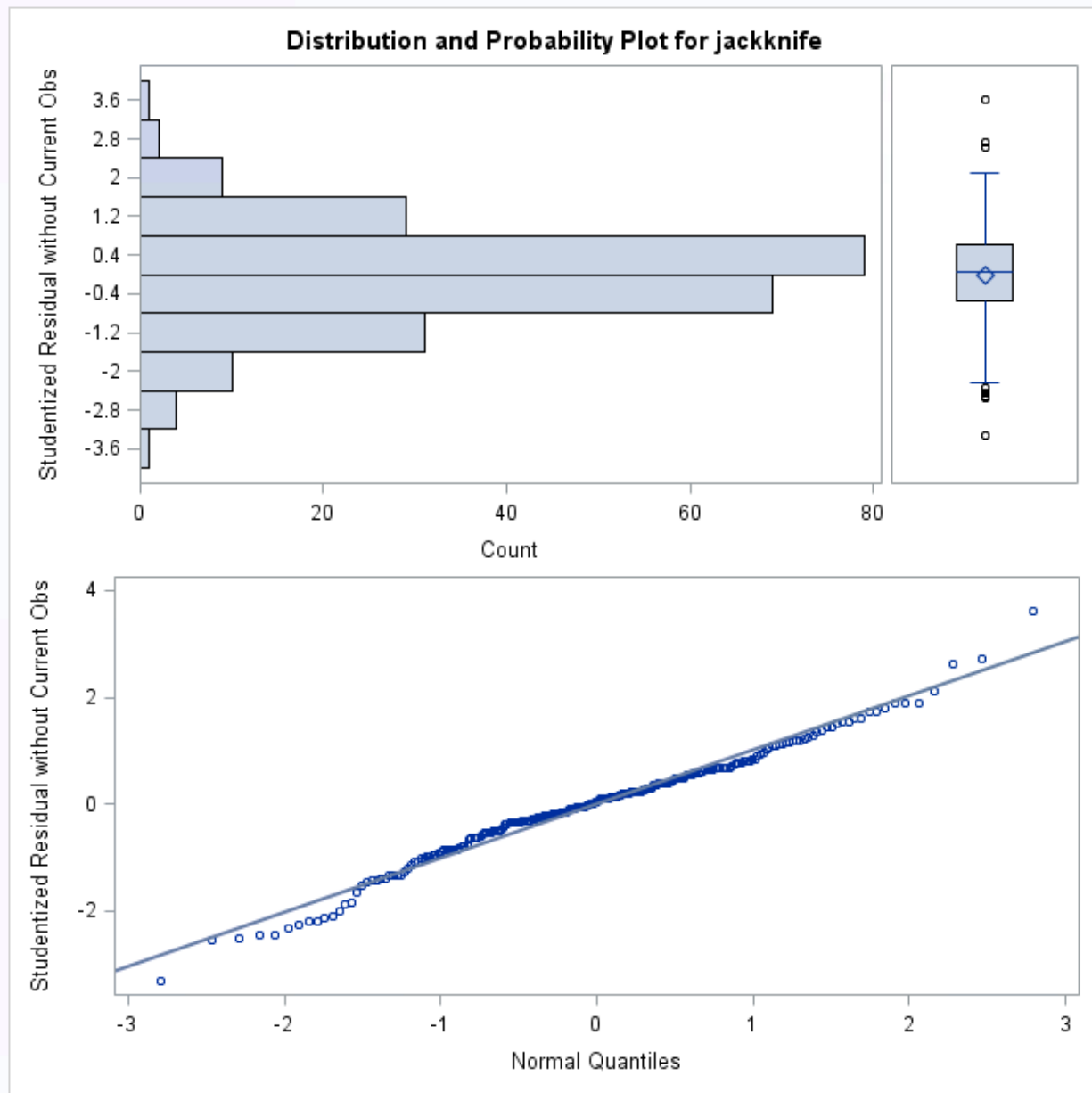
To test if the data follows roughly a normal distribution, we look at a histogram below.

Male Model



In the plot above, the residuals follow a bell curve, so our assumption of normality is met.

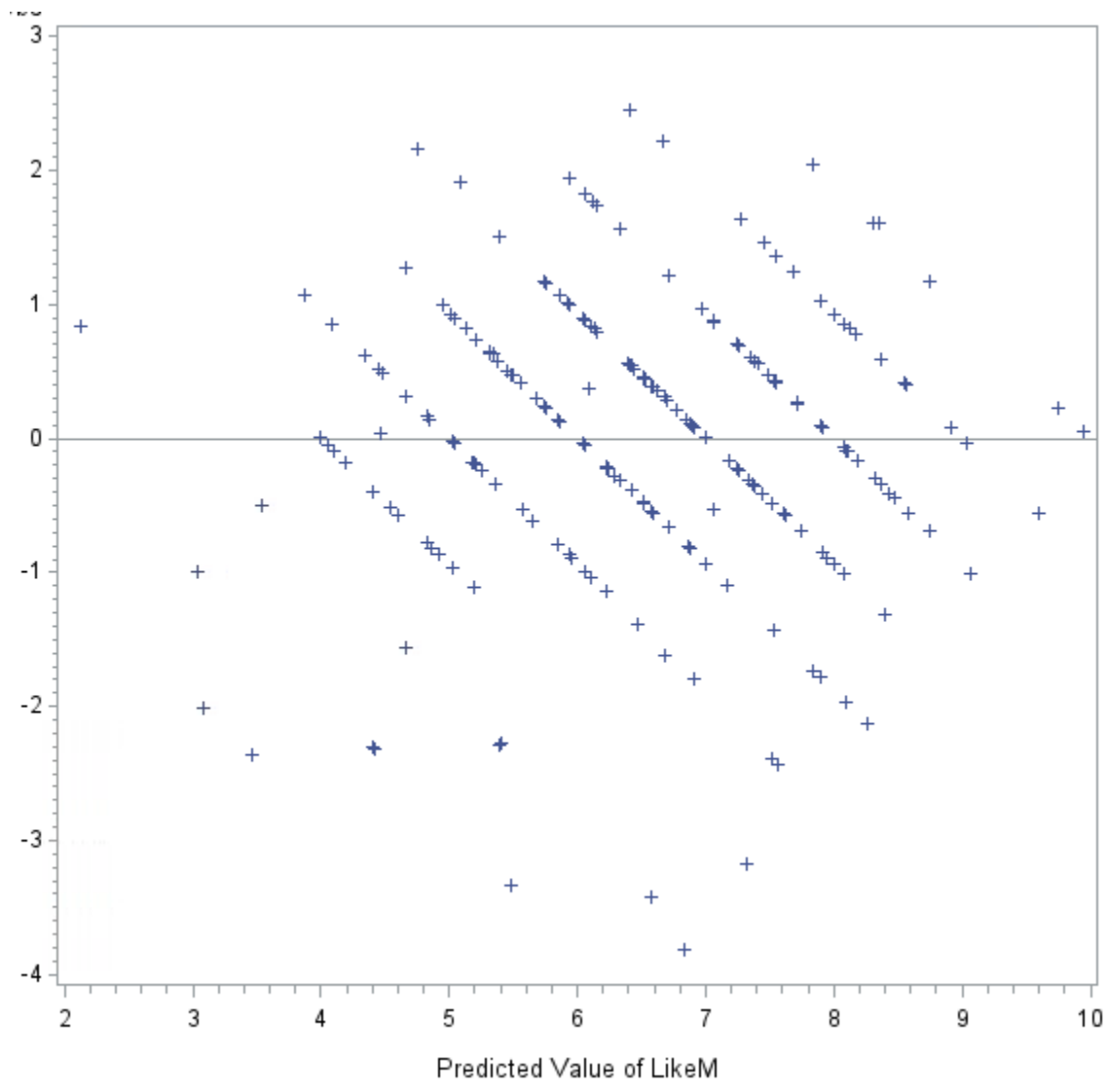
Female Model



In the plot above, the residuals follow a bell curve, so our assumption of normality is met. As the sample mean of the residuals will always be 0, that assumption is met as well for both models.

Next, we'll test for both homogeneity and independence by looking at a plot of the residuals versus predicted values.

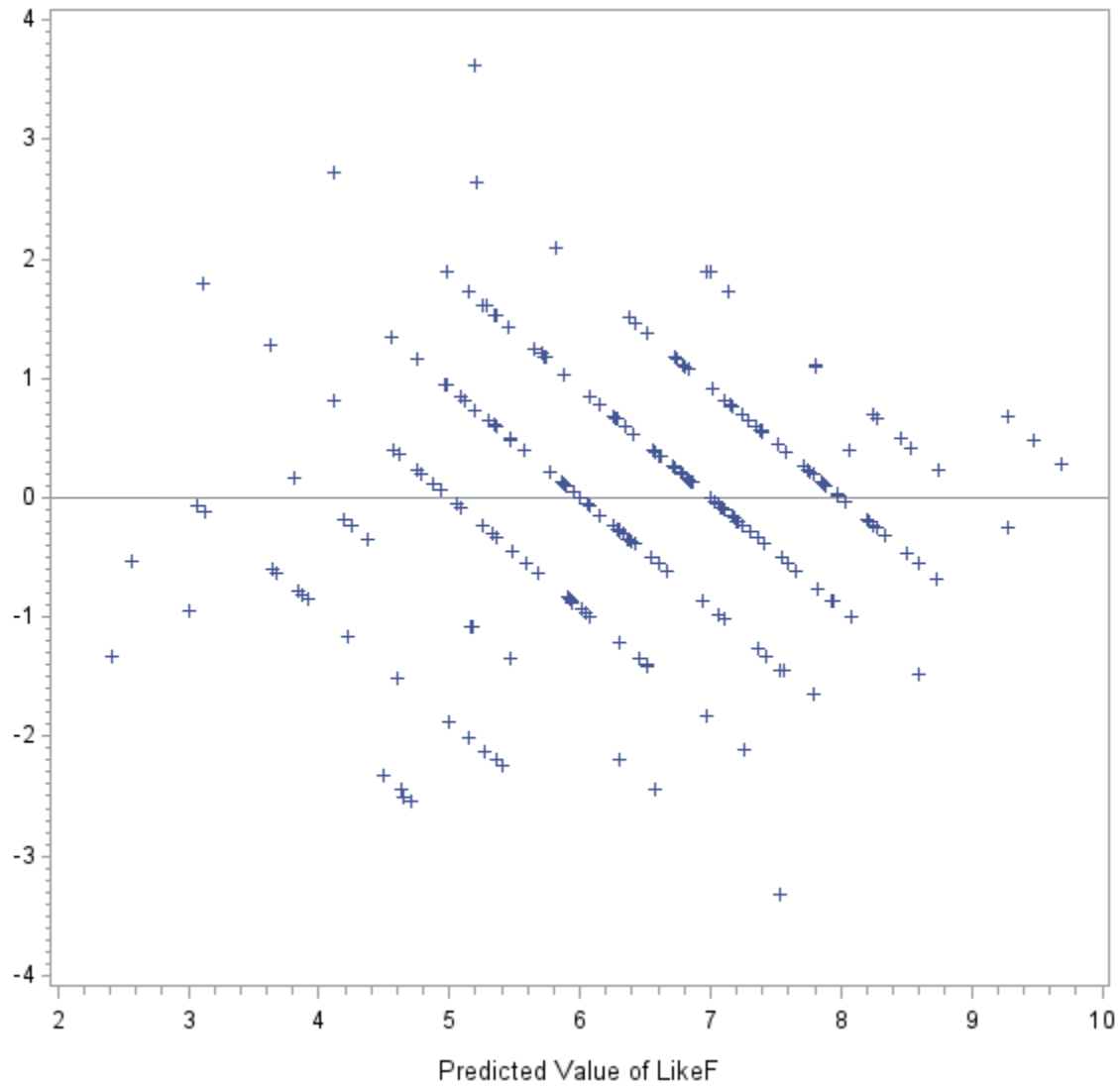
Male Model



As there is no obvious funneling, with random scatter around $y=0$, homogeneity of variance is satisfied. There is a parallel line pattern, which makes sense as our data was discretely scaled 1-10. After considering how data was collected, we can still say independence

is met here, as each observation represents a unique couple, with no individual appearing more than once in the dataset.

Female Model



As there is no obvious funneling, with random scatter around $y=0$, homogeneity of variance is satisfied. Independence is satisfied here as well.

Lastly, we have to check our models for collinearity. If two predictors are too closely related, they are redundant, and we don't need both for prediction. A Variance Inflation Factor greater than 10 is generally seen as problematic, as it's related to a correlation coefficient magnitude of .95 or higher.

Male Model

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	-0.29591	0.41058	-0.72	0.4718	0
AttractiveM	1	0.47522	0.05273	9.01	<.0001	1.83000
FunM	1	0.18607	0.05508	3.38	0.0009	1.90382
SincereM	1	0.18654	0.05671	3.29	0.0012	1.34771
SharedInterestsM	1	0.17600	0.04221	4.17	<.0001	1.40281

Female Model

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	0.16694	0.41701	0.40	0.6893	0
AttractiveF	1	0.28167	0.04528	6.22	<.0001	1.56968
FunF	1	0.25913	0.04877	5.31	<.0001	1.83380
IntelligentF	1	0.20012	0.05391	3.71	0.0003	1.19773
SharedInterestsF	1	0.21188	0.04050	5.23	<.0001	1.56085

Which is not an issue with either of our models. However, VIF is not perfect, so let's check our work with another test, called Eigenanalysis, seen in the figures below.

Male Model

Collinearity Diagnostics (intercept adjusted)						
Number	Eigenvalue	Condition Index	Proportion of Variation			
			AttractiveM	FunM	SincereM	SharedInterestsM
1	2.40542	1.00000	0.06485	0.06469	0.06165	0.06473
2	0.69613	1.85887	0.03393	0.00461	0.63802	0.36054
3	0.53776	2.11495	0.23597	0.17366	0.23288	0.55835
4	0.36068	2.58247	0.66525	0.75704	0.06745	0.01638

Female Model

Collinearity Diagnostics (intercept adjusted)						
Number	Eigenvalue	Condition Index	Proportion of Variation			
			AttractiveF	FunF	IntelligentF	SharedInterestsF
1	2.29497	1.00000	0.07411	0.07490	0.05391	0.07527
2	0.80138	1.69227	0.09661	0.00037561	0.82789	0.06659
3	0.50053	2.14128	0.56746	0.00346	0.00083619	0.70462
4	0.40313	2.38598	0.26183	0.92126	0.11736	0.15352

Considering the largest number in the Condition Index column is less than 30, there does not appear to be any issues with collinearity in either models.

The male and female models differed in their predictors, confirming that each gender weighs characteristics differently – we recommend using both models separately for more accurate pairing. For males, the key drivers of liking were attractiveness, fun, sincerity, and shared interests. Our prediction equation was

$$\text{LikeM} = -0.29591 + 0.47522 * \text{AttractiveM} + 0.18607 * \text{FunM} + 0.18654 * \text{SincereM} + 0.17600 * \text{SharedInterestsM}.$$

The attributes that most significantly predicted how much a female liked her male dating partner were attractiveness, fun, shared interests, and intelligence. The equation I recommend using when predicting how a female will like her male partner is

$$\text{LikeF} = 0.16694 + 0.28167 * \text{AttractiveF} + 0.25913 * \text{FunF} + 0.18654 * \text{IntelligentF} + 0.21188 * \text{SharedInterestsF}.$$

To see whether males and females agree with respect to any of the characteristics of interest, we simply look at the coefficients for variables of the same characteristic. Comparing coefficients across both models, males weight attractiveness more heavily than females — a one unit increase in attractiveness holding all other variables constant increases male liking by 0.47522 versus 0.28167 for females, a difference of about 0.2. I want to reiterate neither couple's being the same race nor close in age were found to be statistically significant in predicting how much an individual liked their partner, for either gender.